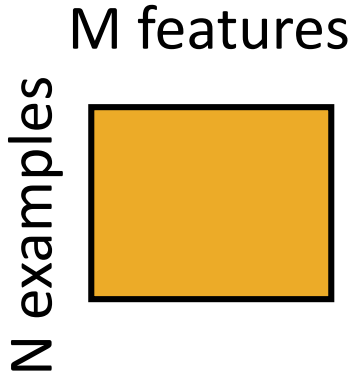
ML Lab

Amlan Chakrabarti

Ensemble Classification- Part 2

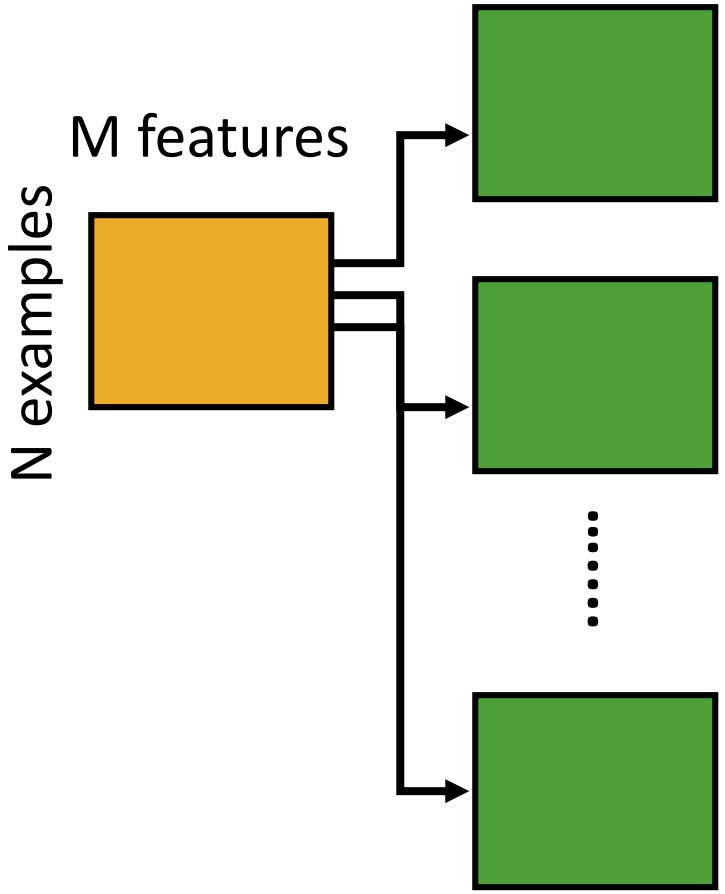
Random Forest Classifier

Training Data

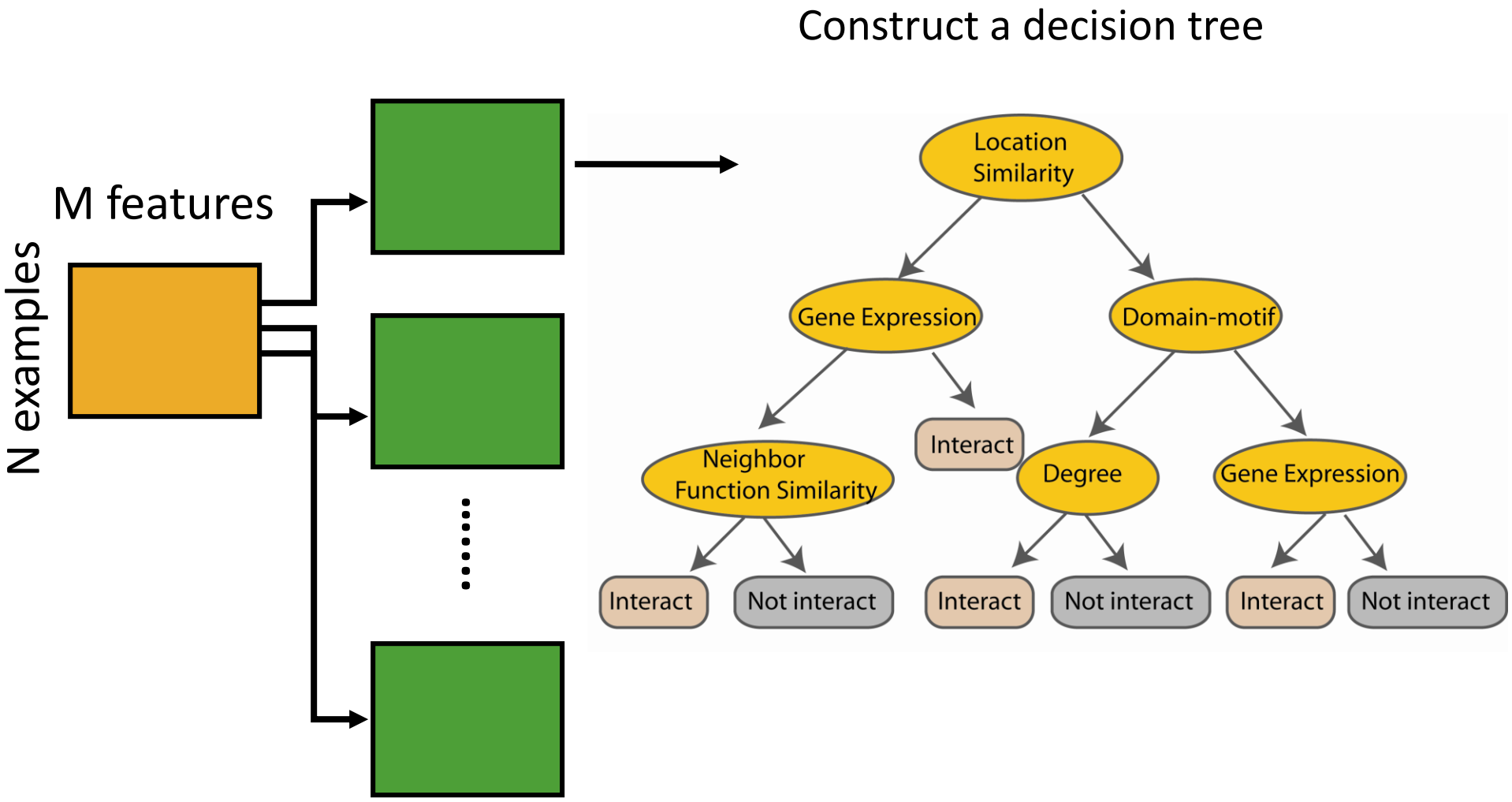


Random Forest Classifier

Create bootstrap samples from the training data

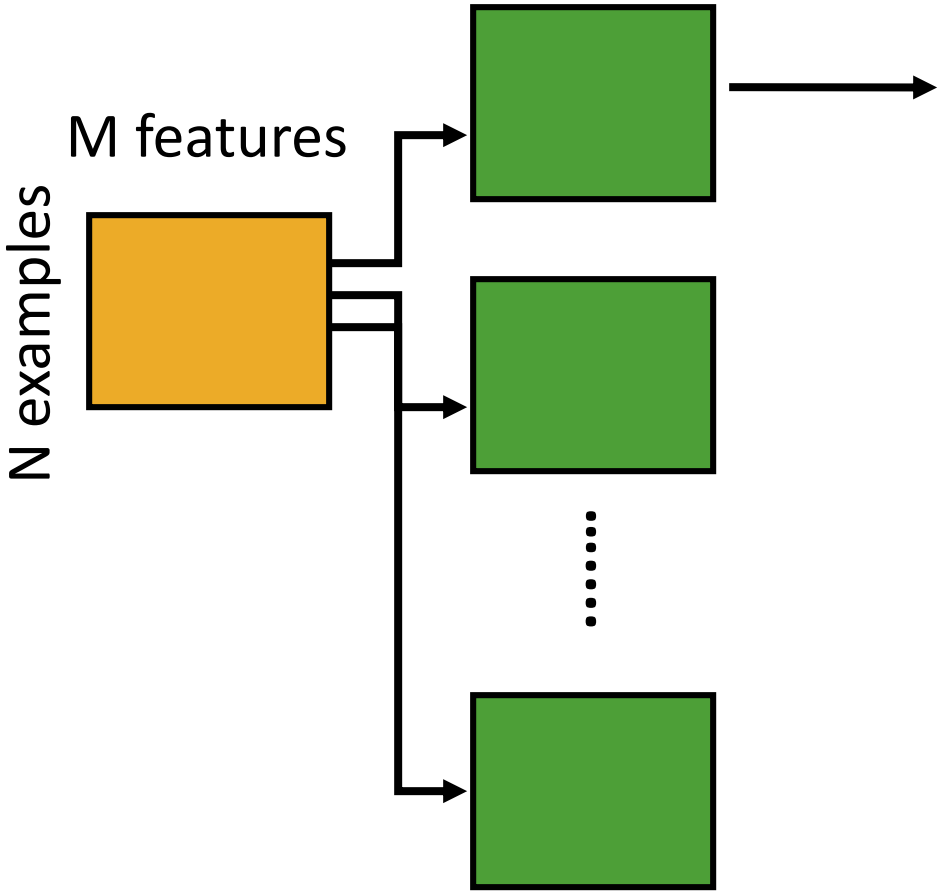


Random Forest Classifier



Random Forest Classifier

- At each node choose the split feature
- choose only among $m<M$ features



How Random Forest actually works (algorithmically)

For **each tree**:

1. Draw a bootstrap sample (data randomness)
2. Grow the tree:
   - **At every node**:
     - Randomly select m features from total M
     - Compute best split using only these m
     - Split on the best of those m

📌 This process is repeated **independently at every node**.

How tree growth works in a Random Forest

Consider one tree inside the Random Forest

Assume:

- Total number of features = **M = 10**
- Number of features tried at each split = **m = 3**

Root node (Tree 1)

- All training samples of this bootstrap dataset reach the root
- **Randomly select 3 features out of 10**

Example: $\{2, 5, 9\}$

- Compute split quality (Gini / entropy) **only for these 3**
- Suppose feature **5** gives the best split

Split on feature 5

Left child node (same tree)

Now consider the **left child node** created by splitting on feature 5.

- Again, **randomly select 3 features** (fresh draw)
- This selection is **independent of the root**

Example: $\{1, 4, 7\}$

- Evaluate splits using only these features
- Suppose feature **7** gives the best split

Split on feature 7

Right child node (same tree)

Now consider the **right child node** of the root.

- Again, **randomly select 3 features**
- Independent random choice

Example: $\{3, 5, 6\}$

- Feature **5 appears again** — this is allowed
- Suppose feature **3** gives the best split

Split on feature 3

This works for multiple trees

What repeats across trees?

- **The procedure repeats**, not the feature sets
- Each tree:
  - Uses a **different bootstrap sample**
  - Makes **different random feature selections at each node**

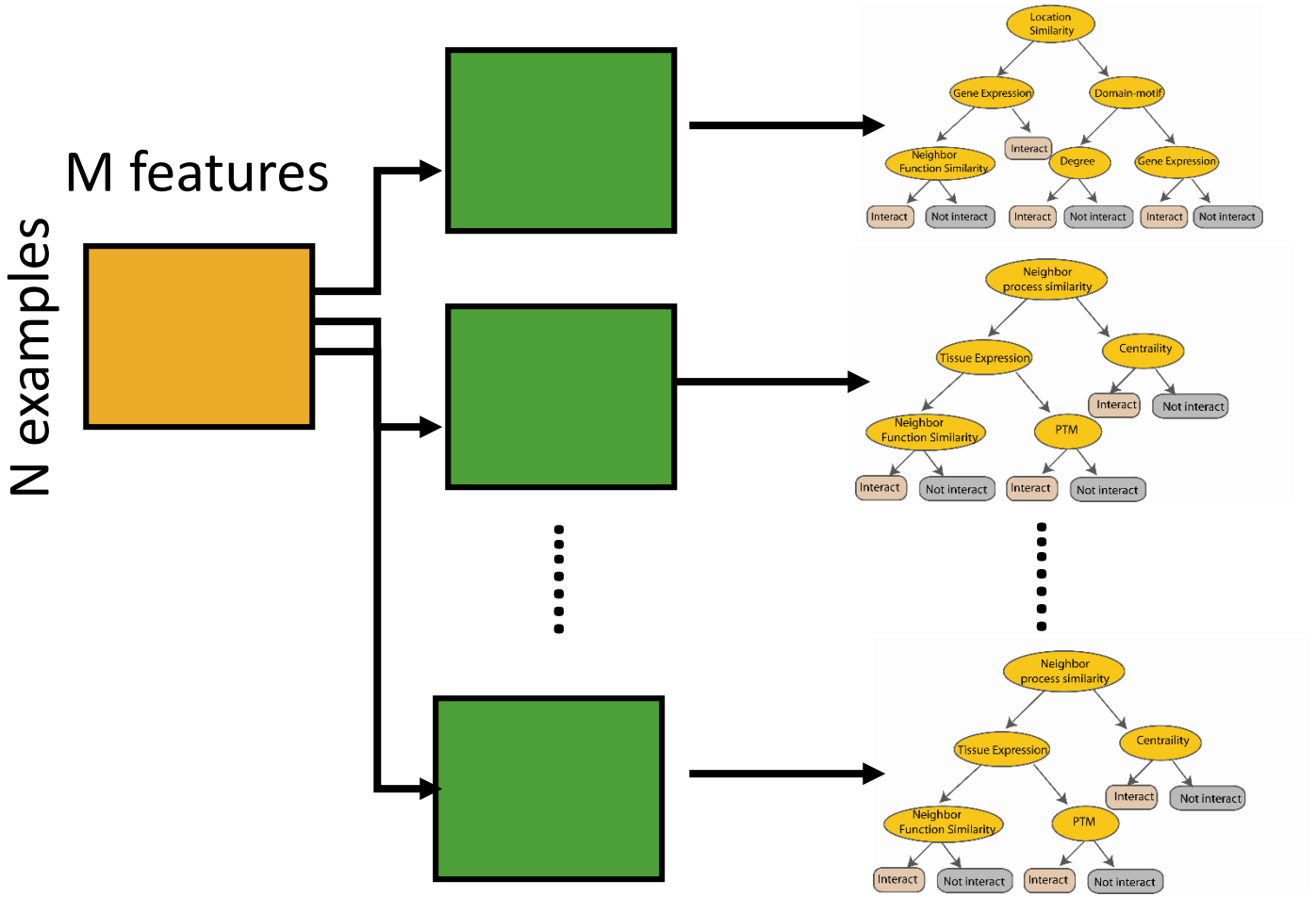
So:

- Tree 1 root might try `{2,5,9}`
- Tree 2 root might try `{1,3,8}`
- Tree 3 root might try `{4,6,10}`

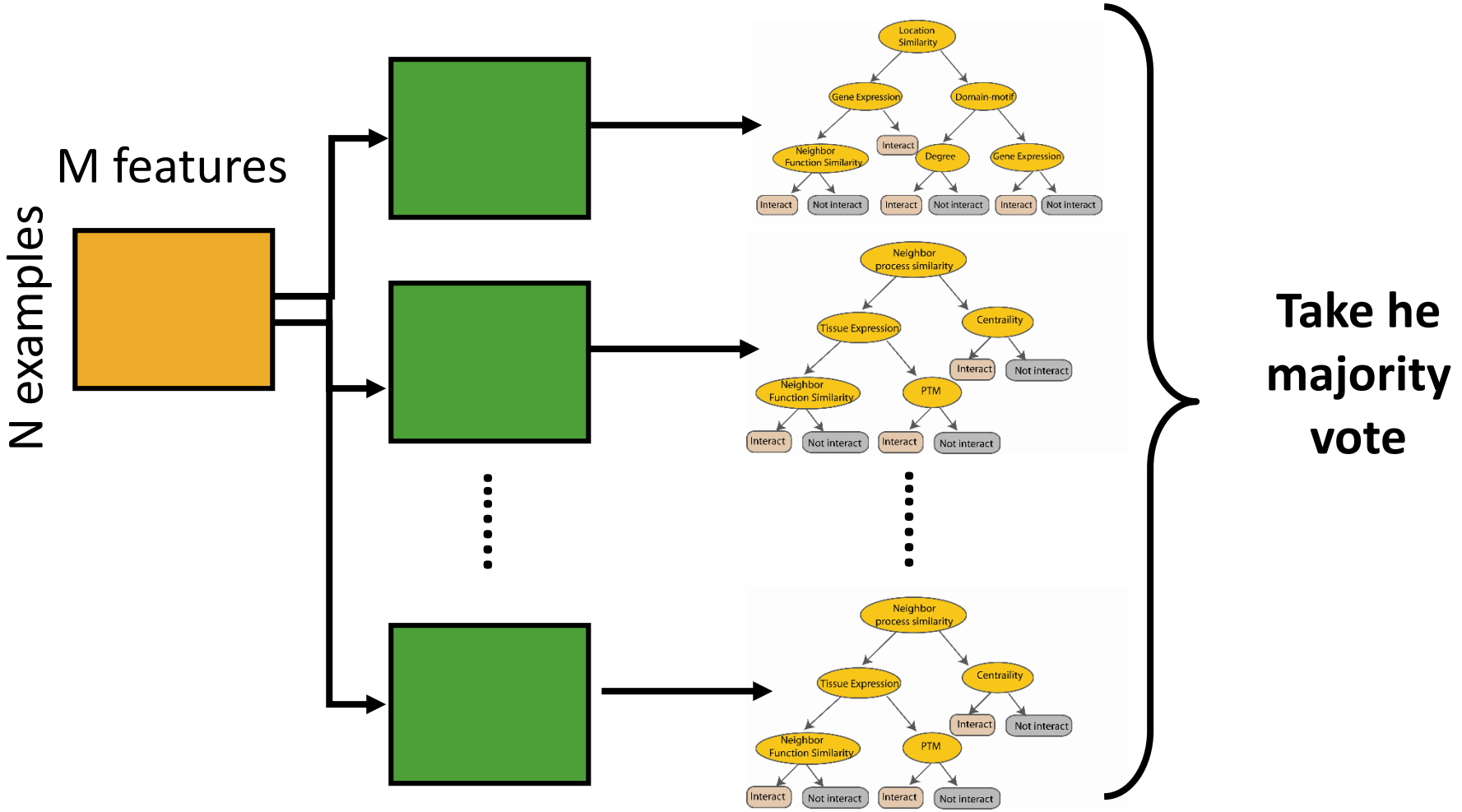
No coordination across trees — randomness is independent.

Random Forest Classifier

Create decision tree from each bootstrap sample



Random Forest Classifier



Methods for Growing the Tree

- **Fix an integer m <= M.**

- **At each node, perform one of the following:**

- **Method 1:**
  1. Choose m attributes randomly,
  2. Compute their information gains,
  3. Choose the attribute with the largest gain to split

- **Method 2:**
  1. Compute the information gain of all M attributes.
  2. Select the top m attributes by information gain.
  3. Randomly select one of the m attributes as the splitting node.



